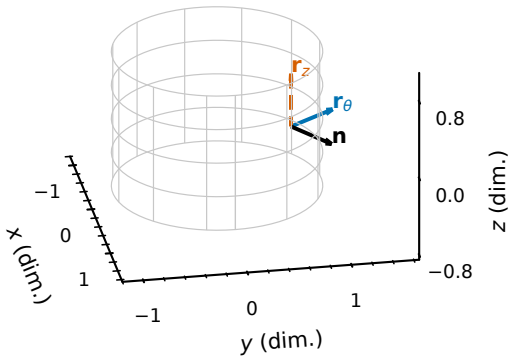
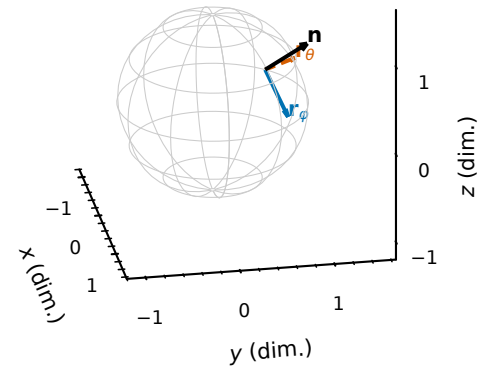
A



$\mathbf{r}_\theta \times \mathbf{r}_z = R\,\mathbf{e}_r$; outward

B



$\mathbf{r}_\varphi \times \mathbf{r}_\theta = R^2 \sin\varphi\,\mathbf{e}_r$; outward